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Conference Chair: Dr. Margaret Loper
Program Chairs: Dr. Dong (Kevin) Jin and Dr. Christopher D. Carothers
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Message from the Conference Chairs

Margaret Loper, Dong (Kevin) Jin, and Christopher D. Carothers

We are delighted to introduce the proceedings of the 2023 edition of the ACM Special Interest Group in Simulation and Modeling International Conference on Principles of Advanced Discrete Simulation (ACM SIGSIM-PADS 2023). This year's program brings together a diverse range of contributions from researchers, practitioners, students, and professionals.

In addition to the traditional double-blind-reviewed technical papers of exceptional quality, we are excited to include several noteworthy elements. An advanced tutorial will delve into emerging topics of interest, such as quantum networking, providing valuable insights for attendees. Furthermore, a colloquium session featuring doctoral students will offer a platform for the presentation and discussion of their research. Additionally, we have incorporated multiple presentations of journal articles recently published in the ACM Transactions on Modeling and Computer Simulation (TOMACS), enriching the program even further.

We express our appreciation to the keynote speaker for this year: Dr. Kalyan Perumalla (Program Manager, U.S. Department of Energy, USA), "Research Opportunities in PDES and AI".

We also express our gratitude to the Best Paper Award Committee: Dr. Wentong Cai (chair), Dr. Peter Barnes, and Dr. Alessandro Pellegrini, for their diligent work to review and select the best paper awards. The candidates short-listed for the best paper award for this year are (in no particular order):

- "Zero Lookahead? Zero Problem. The Window Racer Algorithm," by Philipp Andelfinger, Till Köster, and Adelinde Uhrmacher
- "Workload Interference Prevention with Intelligent Routing and Flexible Job Placement on Dragonfly," by Yao Kang, Xin Wang, and Zhiling Lan.

In keeping with tradition, the best paper award winner from the aforementioned candidates will be announced during the award ceremony session of the conference. This year, we also, for the first time, select the best short paper award, with the objective of encouraging the M&S community to submit short papers describing interesting, novel research that is in an early stage of development or position papers of interest to the community. The best paper award winner of the previous edition of SIGSIM-PADS (2022) was "Temporally synchronized emulation of devices with simulation of networks," by Vignesh Babu and David Nicol. We take this occasion to congratulate the winners.

The current landscape of simulation and modeling, specifically in discrete simulation, presents a range of challenges that also serve as opportunities. These challenges are intertwined with emerging disruptive technologies like quantum technologies, synthetic data generation, realistic virtual worlds, artificial intelligence, and digital twin technologies that rely on simulations as essential components. Simultaneously, the field faces challenges related to attracting and

nurturing a new generation of researchers who can contribute to this vibrant community. We have observed with great enthusiasm that there is a significant amount of ongoing research and development in simulation and modeling, actively pursued by numerous research groups worldwide. These endeavors are aimed at producing outstanding research articles that showcase the advancements and innovation in the field. We eagerly anticipate witnessing the tangible outcomes of this renewed research activity, as it promises to contribute to various aspects of discrete simulation. We anticipate a corresponding increase in publications at ACM SIGSIM-PADS and similar conferences, reflecting the manifestation of this renewed research activity. We remain optimistic about the future and look forward to witnessing the continuous progress and accomplishments within the realm of simulation and modeling as researchers seize the opportunities and tackle the challenges that lie ahead.

Ph.D. Colloquium (Alessandro Pellegrini): This year's edition includes 9 students participating in the Ph.D. Colloquium. The accepted extended abstracts of the colloquium are included in the conference proceeding with the objective of connecting talented young scholars with the established research community in modeling and simulation. A total of 10 extended abstracts were submitted, of which 9 were accepted based on relevance and technical soundness. The accepted students' abstracts showcased research across fields of modeling and simulation and its applications (such as agent-based models, cyber-physical systems, artificial intelligence, and simulation languages) and across countries (including USA, Singapore, Germany, Brazil, and Italy), which demonstrates the international reach of the conference.

Artifact Evaluation (Philipp Andelfinger): ACM SIGSIM-PADS continues its active participation in ACM's Artifact Evaluation initiative, strengthening the trust in published results within the modeling and simulation field. In total, artifacts for four of the accepted papers were submitted for evaluation. We are pleased to announce that after careful evaluation, the reproducibility committee was able to award all of these artifacts the "Results Reproduced" badge. Overall, fourteen out of sixteen possible badges were awarded. We would like to highlight that all four of the artifacts have been made permanently accessible via public archiving sites. Beyond the increased trust through independent reproduction, public availability can accelerate the field's progress by facilitating extensions and comparisons to existing prototypes as well as the reuse of experiment setups. The evaluation process and the final reports demonstrate once again that a swift reproduction of results is achievable but is made possible only through the efforts and diligence on the parts of the authors and independent reviewers. We extend our sincere gratitude to the authors and the reproducibility committee for making this valuable contribution to the community.

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